RAMDEOBABA UNIVERSITY, NAGPUR

Department of Electronics Engineering

CAPSTONE PROJECT PROPOSAL

Course: Deep Learning Lab | Academic Session: 2026–27 | Semester: V

1. Problem Statement and Motivation

Clearly describe the problem to be solved, why it is important, who/what may benefit, and why Deep Learning is suitable. Suggested length: 100–150 words.

Tracking changes in hair while your treatment is going on over time using photographs is difficult because images captured on different days may vary in camera position, head angle, lighting, distance, and background. These variations can make simple image-to-image comparison unreliable and may hide small visual changes in the hair region. The proposed project aims to develop a Deep Learning-based application that helps users capture standardized photographs of their hair from multiple views and compare them consistently over time. A hair-segmentation model will identify the visible hair region in each image, while image-processing techniques will be used to align repeated captures with the user's baseline images. The system will then provide visual and quantitative information about changes in the visible hair region. The application can benefit individuals who want a simple and organized way to monitor changes in their hair appearance through photographs without relying only on subjective visual comparison.

Project Information

Project Title		
Team Member 1	**Name: Aliasgher Husain**	**Roll No.:C/24**
Team Member 2	**Name:Mann Parekh**	**Roll No.:C/29**
Team Member 3	**Name:Shivam Burnwal**	**Roll No.:C/17**
Team Member 4	**Name:Aryan Barhanpure**	**Roll No.:C/26**

2. Project Objectives

State 3–5 clear and measurable objectives. Avoid vague objectives such as "to learn Deep Learning" or "to achieve good accuracy."

1. **To develop and train a Deep Learning-based image segmentation model capable of identifying the visible hair region from input photographs.**
2. **To develop a standardized image-capture and preprocessing pipeline for front, left 45°, and right 45° views**
3. **To align subsequent photographs with the corresponding baseline image and perform consistent hair-region comparison.**
4. **To calculate quantitative measures of visible hair-region coverage and change between baseline and subsequent captures**
5. **To evaluate the proposed system using segmentation metrics and compare its performance with a traditional computer-vision-based alignment approach.**

3. Dataset

Provide the dataset name/source, size, type of data, number of classes/outputs, and briefly mention preprocessing requirements and anticipated issues.

Dataset Name	CelebAMask-HQ
Dataset Source	CelebAMask-HQ, The Chinese University of Hong Kong and associated authors.
Dataset Size	30,000 high-resolution face images with pixel-level facial-component annotations. The dataset includes a dedicated hair class.
Type of Data	Images with semantic segmentation masks
No. of Classes / Outputs	Binary hair segmentation for the proposed task — Hair / Non-Hair.

4. Proposed Methodology

Insert a simple block diagram / flowchart / system architecture. Briefly explain the major stages and justify the selected Deep Learning model/architecture.

Camera Input → Preprocessing → Deep Learning Hair Segmentation → Image Alignment → Hair-Region Comparison → Quantitative Analysis → Progress History

Proposed Model / Architecture:

The proposed system will first capture photographs from standardized front, left 45°, and right 45° views. The input images will be resized and normalized before being passed to a U-Net-based segmentation model. The model will generate a pixel-level mask identifying the visible hair region. After segmentation, the captured image will be aligned with the corresponding baseline photograph using feature-based image alignment. The aligned hair masks will then be compared to estimate changes in visible hair-region coverage. The system will present both a visual before/after comparison and quantitative change information through a simple application interface. A U-Net architecture is proposed because the project requires pixel-level segmentation rather than only image-level classification. Its encoder-decoder structure is suitable for learning both high-level image features and detailed boundaries of the hair region. The existing traditional image-alignment implementation will also be retained as a baseline for comparison with the proposed Deep Learning-based pipeline.

Brief justification:

U-Net is suitable because the project requires pixel-level identification of the hair region. Its encoder-decoder architecture can capture both important image features and fine hair boundaries, making it appropriate for accurate hair segmentation.

5. Experimental and Evaluation Plan

Mention the train/validation/test split, loss function, optimizer, appropriate evaluation metrics, and any baseline/comparison model.

Data Split	**Training: 80% | Validation: 10% | Testing: 10%**
Loss Function	Binary Cross-Entropy combined with Dice Loss
Optimizer	Adam
Evaluation Metrics	Dice Score, Intersection over Union (IoU), Precision, Recall, and validation/test loss.

Baseline / Comparison	Traditional feature-based alignment using AKAZE feature matching and RANSAC-based homography will be used as the initial computer-vision baseline. The proposed Deep Learning segmentation pipeline will then be evaluated and compared through quantitative results and qualitative visual comparisons.

6. Expected Outcome / Deliverables

State what will be demonstrated at the end of the project: trained/evaluated model, comparative analysis, prototype/application (if applicable), and analysis of limitations/failure cases.

- *A trained and evaluated Deep Learning model for visible hair-region segmentation.*
- *A preprocessing and image-alignment pipeline for repeated hair photographs.*
- *A working application for capturing baseline and subsequent photographs from multiple views.*
- *Visual before/after comparison of hair regions.*
- *Quantitative analysis of visible hair-region changes across photographs.*
- *Experimental comparison between traditional computer-vision alignment and the proposed Deep Learning-based approach.*
- *Analysis of model limitations and common failure cases*

7. Work Distribution

Indicate the major responsibility of each member. Every member must understand the complete project.

Sr. No.	Team Member	Major Responsibility
1	Aliasgher Husain	Model training, evaluation and experimentation
2	Mann Parekh	Dataset preparation, preprocessing and Deep Learning model development
3	Shivam Burnwal	Backend, image alignment and quantitative comparison
4	Aryan Barhanpure	Frontend application, integration, testing and demonstration

8. Proposed Timeline

Students may modify the phases according to the nature of the project.

Phase	Proposed Activity
Phase 1	Problem finalization, literature review and dataset preparation
Phase 2	Dataset preprocessing, annotation conversion and traditional baseline implementation
Phase 3	U-Net model development, training and validation
Phase 4	Model improvement, augmentation and application-specific testing
Phase 5	Integration, quantitative evaluation and error analysis

Phase 6	Final demonstration, project report and presentation

9. References

Provide 3–6 relevant references. Use a consistent citation style, preferably IEEE. Include research papers, dataset sources, official documentation, books, or reputable technical resources.

[1] C.-H. Lee, Z. Liu, L. Wu, and P. Luo, "MaskGAN: Towards Diverse and Interactive Facial Image Manipulation," IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020. The associated CelebAMask-HQ dataset provides 30,000 high-resolution images with manually annotated facial-component masks, including hair

[2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2015.

[3] CelebAMask-HQ Dataset, The Chinese University of Hong Kong, official dataset resource. CelebAMask-HQ Dataset

[4] OpenCV Documentation, "Feature Detection and Description / Homography," OpenCV technical documentation.

Submission Guidelines

- The complete proposal, including references, must not exceed 3 pages.
- The proposal must clearly communicate the problem, dataset, proposed methodology, and evaluation plan.
- Use diagrams, flowcharts, and tables where they improve clarity; avoid unnecessary theoretical descriptions.
- All external datasets, figures, architectures, papers, and technical resources must be appropriately acknowledged.
- Every team member must be prepared to explain the complete proposal.
- **Students may present their project proposal using a PowerPoint Presentation (PPT) along with this template.**
- The PPT should broadly follow: Problem → Objectives → Dataset → Methodology → Evaluation → Expected Outcome → References.